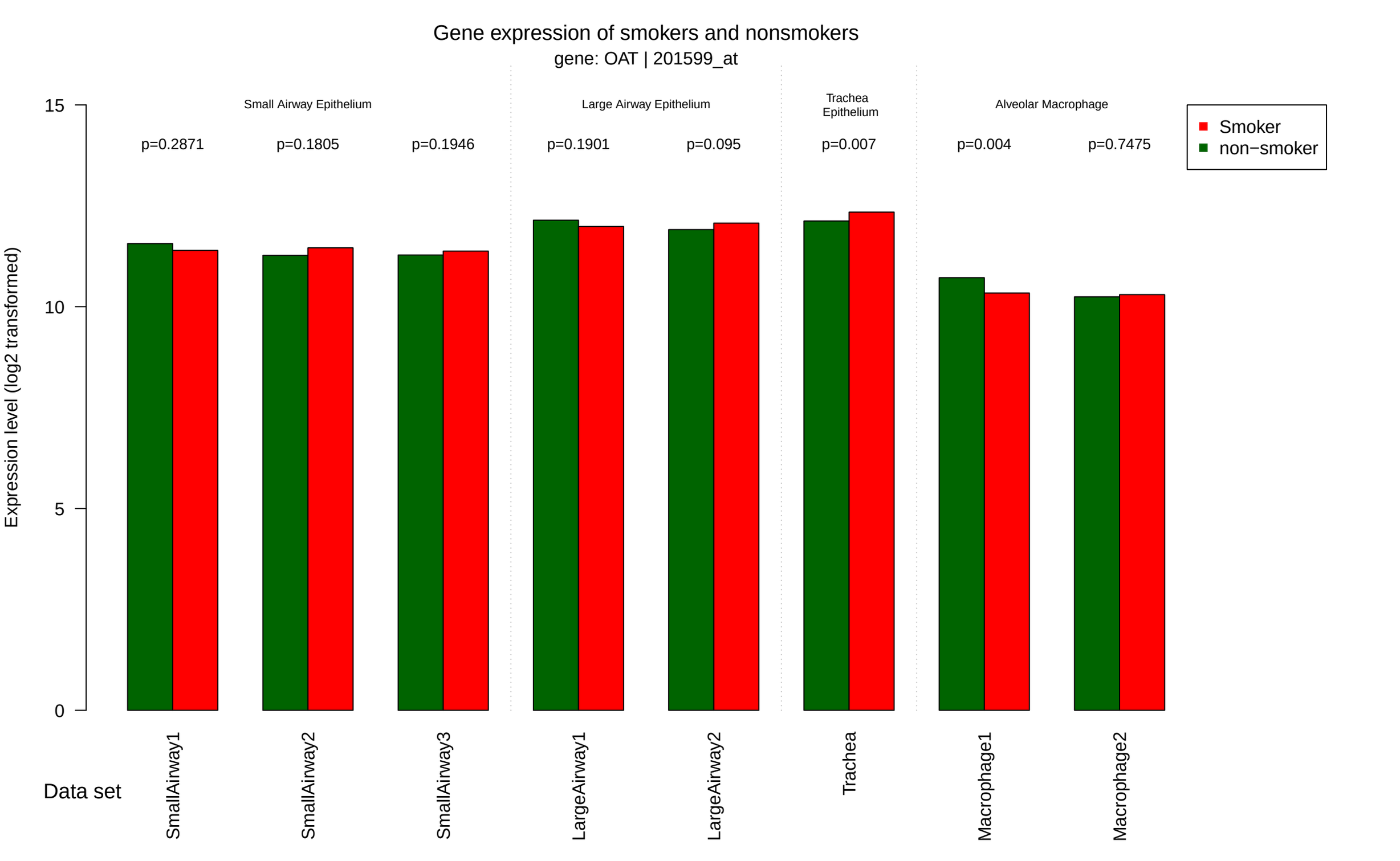
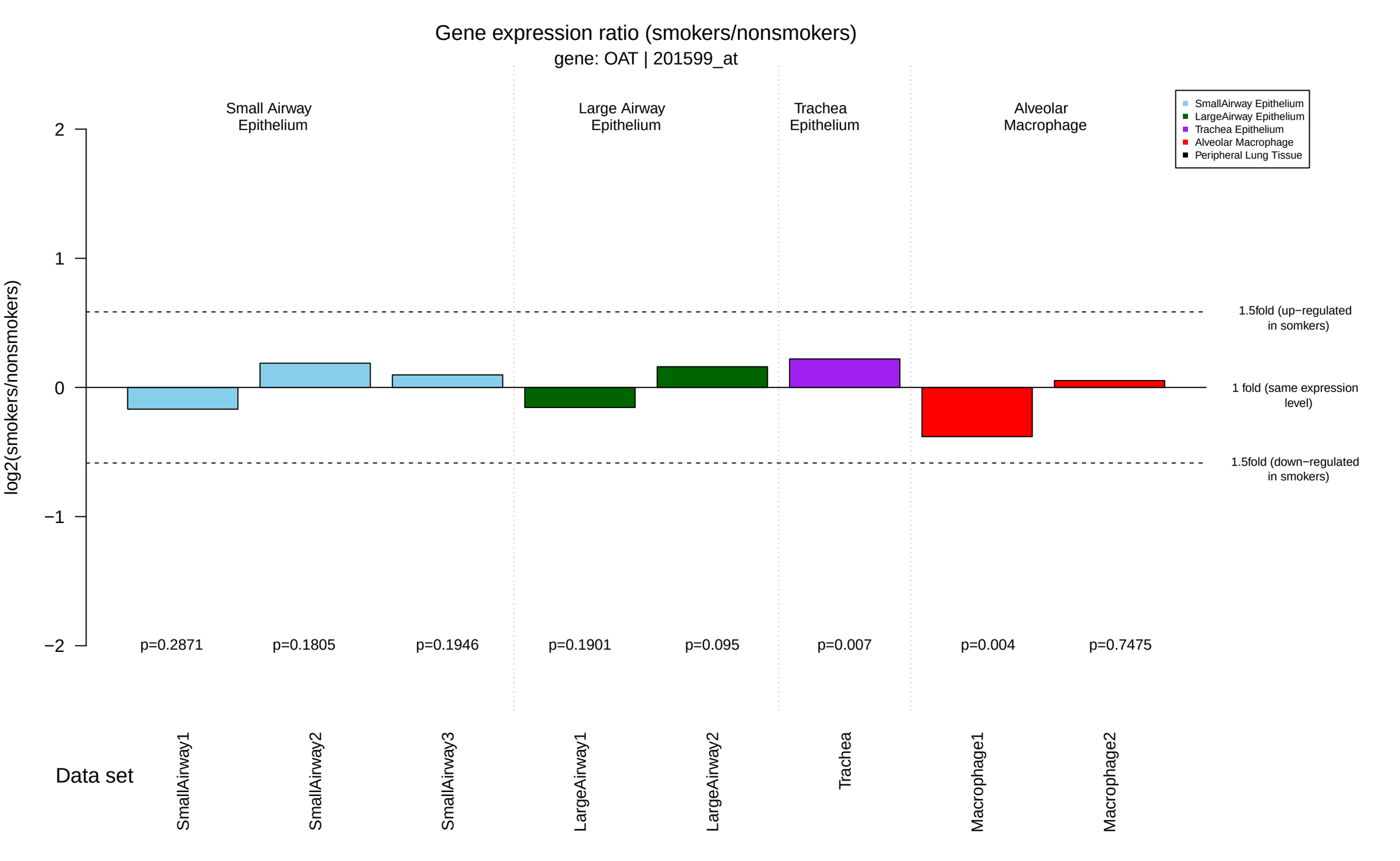
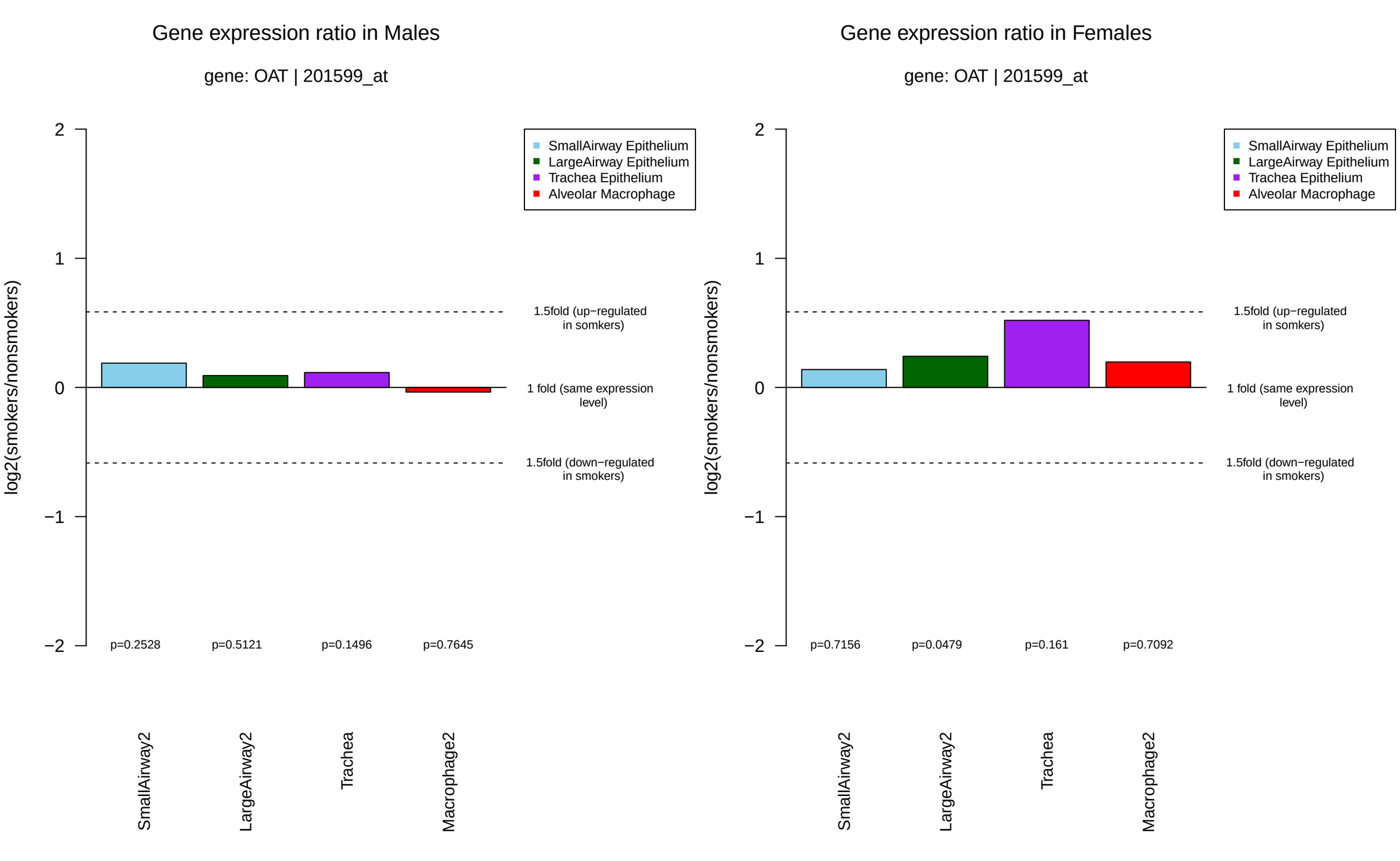
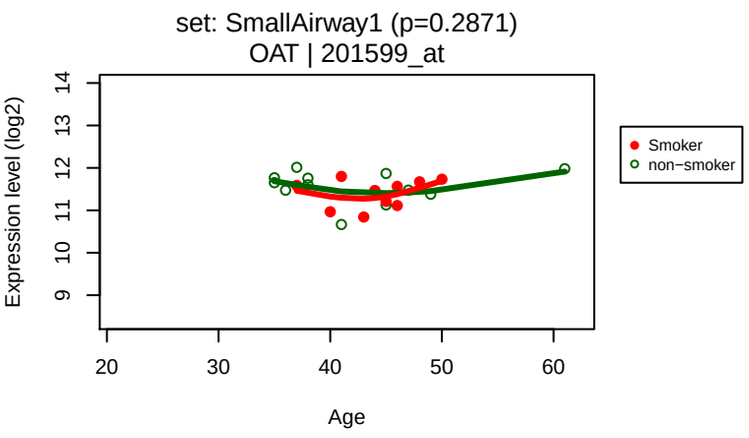
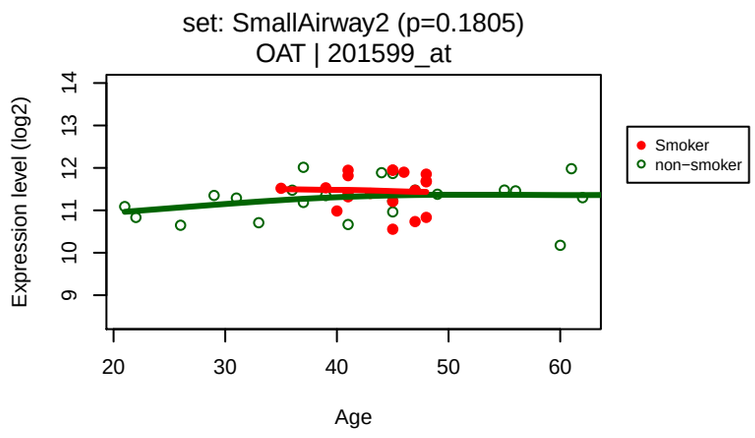
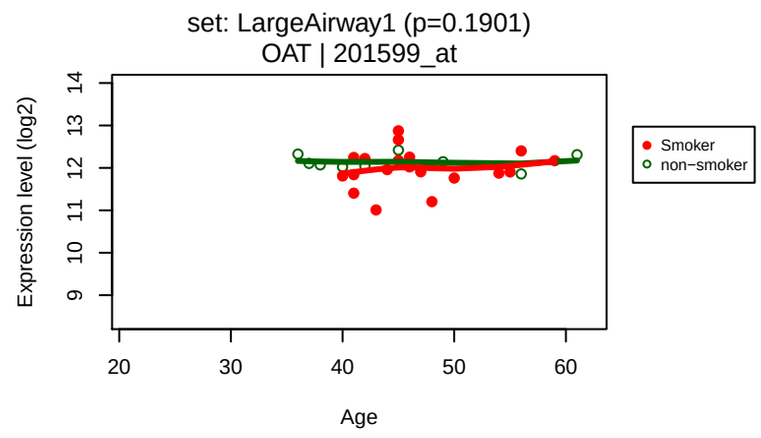

Gene expression of smokers and nonsmokers
gene: OAT | 201599_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2871
p=0.1805
p=0.1946
p=0.1901
p=0.095
p=0.007
p=0.004
p=0.7475
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

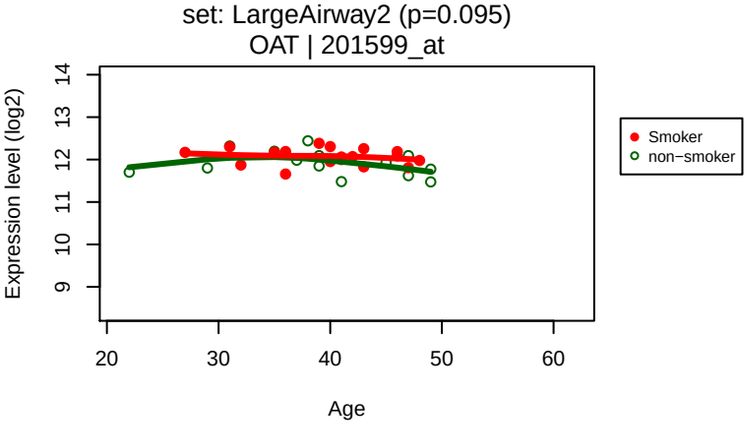

Gene expression ratio (smokers/nonsmokers)
gene: OAT | 201599_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2871
p=0.1805
p=0.1946
p=0.1901
p=0.095
p=0.007
p=0.004
p=0.7475
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

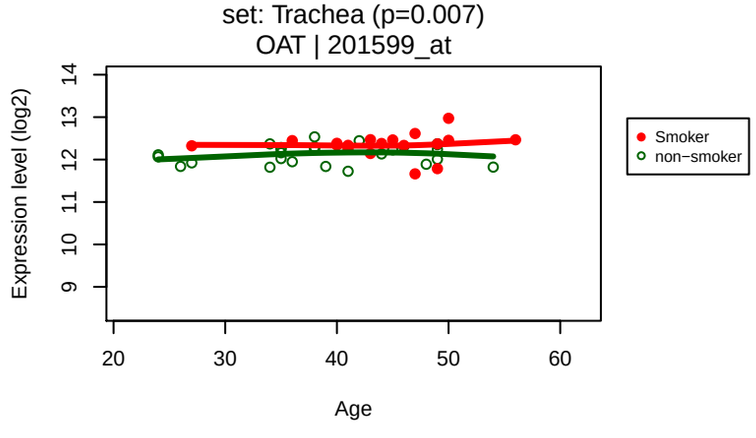

Gene expression ratio in Males
gene: OAT | 201599_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2528
p=0.5121
p=0.1496
p=0.7645
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OAT | 201599_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7156
p=0.0479
p=0.161
p=0.7092
SmallAirway2
LargeAirway2
Trachea
Macrophage2

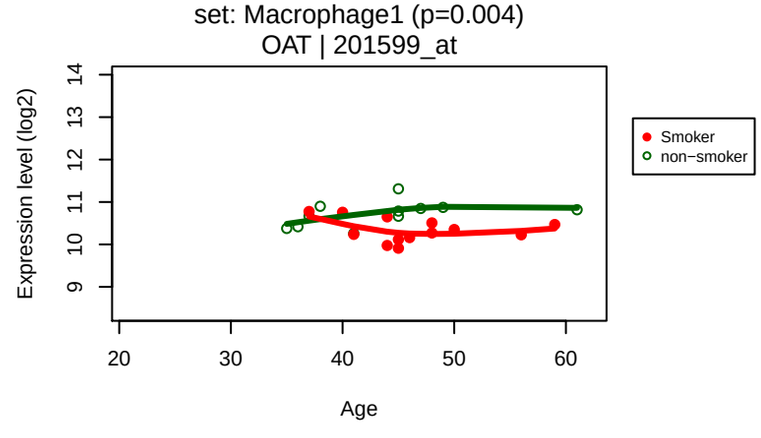

set: SmallAirway1 (p=0.2871)
OAT | 201599_at
Expression level (log2)
Age
Smoker
non−smoker

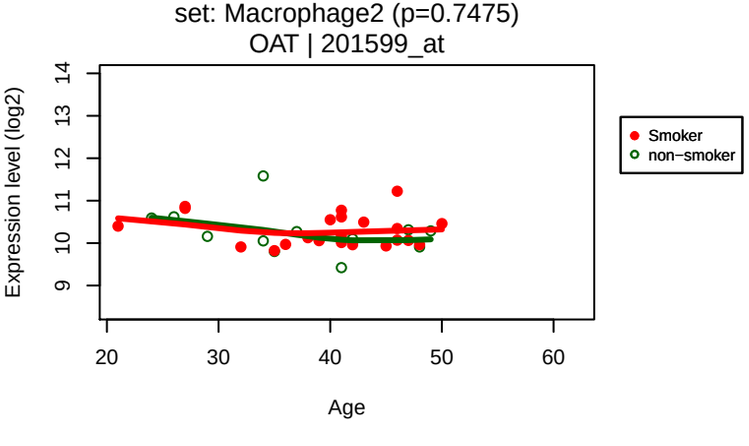

set: SmallAirway2 (p=0.1805)
OAT | 201599_at
Expression level (log2)
Age
Smoker
non−smoker

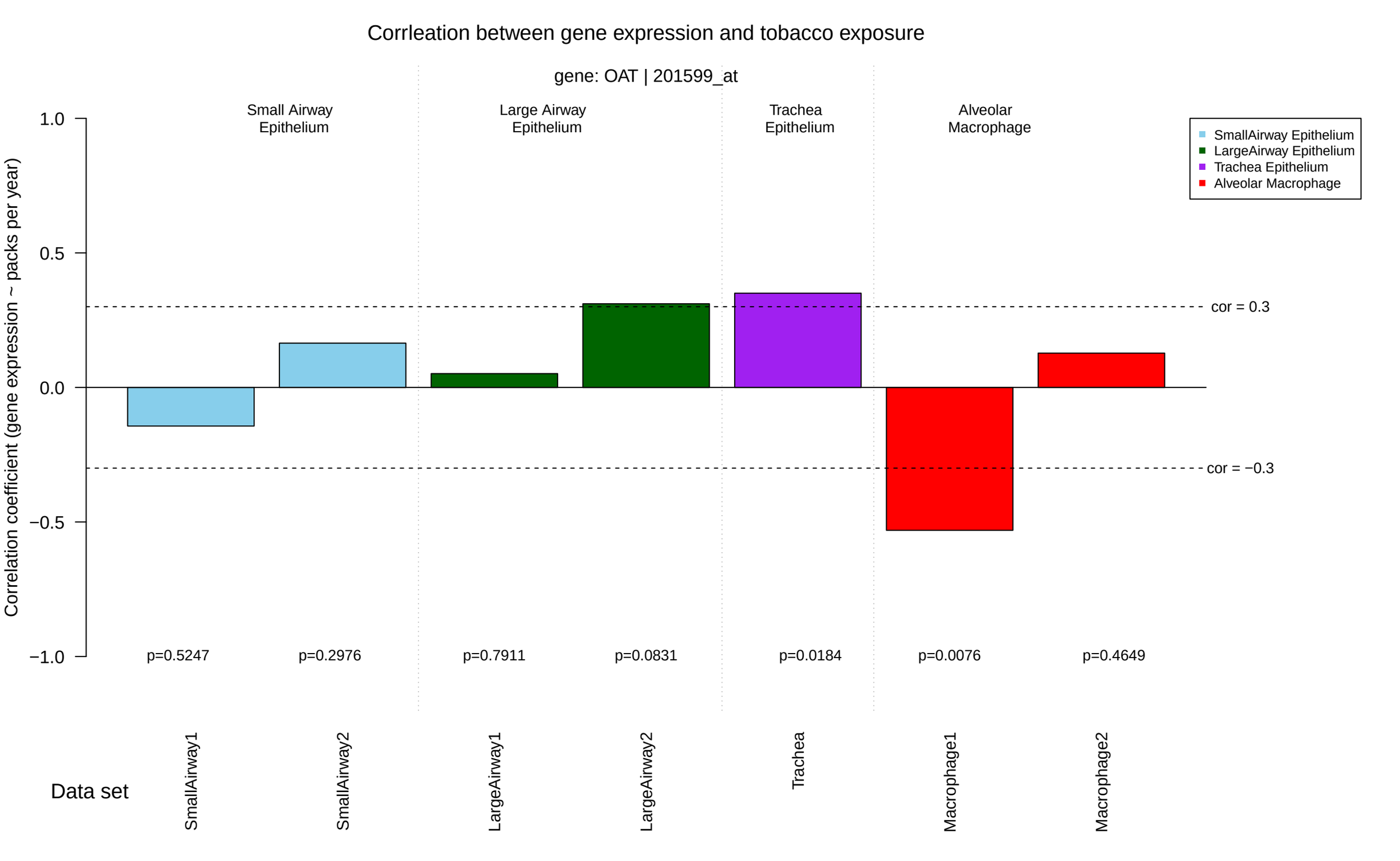

set: LargeAirway1 (p=0.1901)
OAT | 201599_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.095)
OAT | 201599_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.007)
OAT | 201599_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.004)
OAT | 201599_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7475)
OAT | 201599_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OAT | 201599_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5247
p=0.2976
p=0.7911
p=0.0831
p=0.0184
p=0.0076
p=0.4649
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2